

Phase 1: Project Planning & Management

Project Title: AI-Based In-Cabin Monitoring System

Program: DEPI (روداد مصر الرقمية) - AI & Data Science Diploma

Track: Graduation Project

1. Project Proposal

Overview

The AI-Based In-Cabin Monitoring System is an intelligent, real-time edge-computing solution designed to drastically improve vehicle safety and security. Utilizing advanced Artificial Intelligence, Deep Learning, and Computer Vision technologies, the system continuously monitors the vehicle cabin to analyze driver and passenger behavior. Designed to operate completely offline on consumer-grade hardware (NVIDIA GTX 1650 / 1660 Super) with CUDA acceleration, it ensures real-time performance and absolute privacy.

Objectives

- **Improve Driver Safety:** Actively monitor and alert drivers in real-time to prevent accidents caused by fatigue, microsleeps, or visual distraction.
- **Enhance Vehicle Security:** Implement robust face authentication with anti-spoofing to restrict vehicle operation to authorized personnel only.
- **Detect Threats:** Identify violent activities and brandished weapons within the cabin to protect passengers and drivers.
- **High-Performance Edge Computing:** Achieve high detection accuracy ($\geq 90\%$) and ultra-low response latency (≤ 2 seconds) without relying on cloud processing.

Scope and Target Audience

The system integrates four distinct AI modules (Drowsiness, Face Recognition, Distraction, and Violence Detection) unified under a local Flask-based dashboard.

Target Stakeholders include:

- Fleet Management Companies
- Ride-sharing Services (e.g., Uber, Careem)

- Public Transportation Operators
- Armored Vehicles and Cash Transport Services
- Police Patrol Units

2. Project Plan & Milestones

The project spans approximately 16 weeks, from conceptualization to final deployment.

Timeline (Gantt Chart Summary)

Phase	Duration	Milestones & Deliverables
Project Planning	Feb 1 – Feb 10, 2026	Define objectives, scope, roles, and initial timeline.
Literature Review	Feb 10 – Feb 25, 2026	Research existing DMS solutions, YOLO architectures, and temporal models (MoViNet).
Requirements Gathering	Feb 20 – Mar 5, 2026	Finalize functional/non-functional constraints, stakeholder needs, and datasets.
System Analysis & Design	Mar 1 – Mar 20, 2026	Draft UML diagrams, hardware pipeline architectures, and database schemas.
AI Model Development	Mar 10 – Apr 20, 2026	Train YOLOv8n, YOLO11n, ArcFace, and VD-MIL models; refine datasets.
System Integration	Apr 15 – May 5, 2026	Merge sub-systems into unified Flask backend and

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		resolve thread-safety issues.
Testing & Optimization	May 1 – May 15, 2026	Run live inference testing, threshold tuning, and VRAM optimization.
Documentation & Delivery	May 10 – May 20, 2026	Finalize GitHub repository, READMEs, technical documents, and presentation.

3. Task Assignment & Roles

The system's modular architecture allows for parallel development. Responsibilities are divided as follows:

Team Member	Role	Core Responsibilities
Omar Moselhy	Team Leader & AI Engineer	Project management, system architecture integration, and development of the Violence & Weapon Detection module (YOLOv8n + VD-MIL).
Aya Sabry	AI Engineer	Development of Face Recognition (ArcFace/InsightFace) with anti-spoofing, and Distraction Detection (YOLOv8n).

Team Member	Role	Core Responsibilities
Mostafa Essam	AI Engineer	Development of the Drowsiness Detection module utilizing a Hybrid Vision Ensemble (YOLO11n + MediaPipe FaceMesh) and SQLite logging.

4. Risk Assessment & Mitigation Plan

Identified Risk	Impact	Mitigation Strategy
Integration Complexity	High	Standardized on Python 3.11.5, PyTorch, and a unified Flask backend. Module boundaries are clearly defined to prevent dependency conflicts (e.g., using threading locks for global states).
Real-Time Performance Bottlenecks	High	Utilized nano-tier models (YOLOv8n, YOLO11n) and efficient backbones (MoViNet A0). Ensured hardware acceleration via CUDA on GTX 1650/1660 Super.
Varying Lighting & Environmental Conditions	Medium	Applied heavy dataset augmentation (HSV shifts, random erasing) mimicking night driving, glare, and varied cabin lighting. Face recognition incorporates

Identified Risk	Impact	Mitigation Strategy
		robust anti-spoofing.
Dataset Quality & Class Imbalance	Medium	Migrated from noisy general datasets to highly focused, balanced sets (e.g., discarding EfficientNet for YOLOv8 in distraction, upgrading from 455 to 6,803 images for drowsiness).

5. Key Performance Indicators (KPIs)

To ensure the system meets commercial and safety standards, the following baseline metrics must be achieved:

- **Detection Accuracy:** $\geq 90\%$ across all primary models (mAP@50 or classification accuracy).
- **System Response Time:** ≤ 2 Seconds from event occurrence to visual/audio dashboard alert.
- **Inference Latency:** Sub-50ms per frame to maintain real-time capabilities.
- **Frame Rate:** Sustain continuous operation at ~30 FPS under full system load.
- **System Stability:** 100% offline continuous operation without memory leaks or thermal throttling on target GPUs.